Figure 3D · Measured barcode changes

Measured class	Observed source barcode	Selected target barcode
Corrected off-list	Outside the full chemistry catalog	In the tool-selected set
Failure Mode 1	In the chemistry catalog; outside the matched filtered reference	In the tool-selected set
Failure Mode 2	In the matched filtered reference	In both the matched filtered reference and tool-selected set

Every class requires an observed change from source to target. Classes may overlap; source-set membership does not establish biological origin.